

Bug Detection (Defect Detection) using Generative AI & GitHub Copilot

What is Bug Detection?

Bug Detection is the process of **identifying defects, errors, or inconsistencies** in software that cause incorrect or unexpected behavior.

A **bug** occurs when:

- Software does not meet requirements
 - Produces incorrect output
 - Crashes or behaves abnormally
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Bug Detection – Importance

Area	Why It Matters
Quality Assurance	Ensures stable and reliable product
Cost Saving	Early detection reduces fix cost
User Satisfaction	Prevents poor user experience
Security	Detects vulnerabilities
Compliance	Meets industry standards

Cost of fixing a bug after production can be **10x–100x higher** than during development.

Bug Detection – Common Methods

1. Static Testing (Before Execution)

- Code review
- Static code analysis
- Linting

2. Dynamic Testing (During Execution)

- Unit Testing
- Integration Testing
- System Testing
- UAT

3. Manual Bug Detection

- Exploratory testing
- Checklist-based testing

4. Automated Bug Detection

- Selenium, Cypress
 - Unit frameworks (JUnit, PyTest)
 - CI pipelines
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Generative AI in Bug Detection

Generative AI analyzes:

- Source code
- Test cases
- Logs
- Historical defects

and predicts or identifies **potential bug patterns**.

How GenAI Helps

Finds logical errors
Detects edge cases
Identifies risky code
Suggests fixes
Generates tests to expose bugs

Generative AI in Bug Detection – Benefits

Benefit	Description
Faster Detection	Seconds instead of hours
Higher Coverage	Finds hidden paths
Reduced Manual Effort	Less repetitive work
Early Bug Discovery	Shift-left testing
Better Code Quality	Prevents defects

Bug Detection using GitHub Copilot – Key Features

- Real-time bug highlighting
 - Suggests safer code alternatives
 - Generates missing validations
 - Writes unit tests
 - Explains suspicious logic
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DEMO: Bug Detection using GitHub Copilot

Scenario:

Login API in Node.js with a bug.

Buggy Code

```
app.post("/login", (req, res) => {
  const { email, password } = req.body;

  if(email == "" || password == "") {
    res.send("Missing fields");
  }

  if(email == "admin@test.com" && password == "1234") {
    res.send("Login successful");
  } else {
    res.send("Invalid credentials");
  }
});
```

Problems

- No return after missing fields
 - Password in plain text
 - No status codes
 - No input validation
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Ask Copilot Prompt

Analyze this login API and identify bugs and security issues.
Suggest corrected code.

Copilot Suggested Fix

```
app.post("/login", (req, res) => {
  const { email, password } = req.body;

  if (!email || !password) {
    return res.status(400).json({ message: "Email and password required" });
  }

  if (email === "admin@test.com" && password === "1234") {
    return res.status(200).json({ message: "Login successful" });
  }

  return res.status(401).json({ message: "Invalid credentials" });
});
```

Copilot Additional Suggestions

```
// Add password hashing
// Add rate limiting
// Add input sanitization
```

DEMO 2: Copilot Generating Tests to Catch Bugs

Prompt:

Generate Jest test cases to validate login API.

Generated Test

```
test("should fail if email missing", async () => {
  const res = await request(app)
    .post("/login")
    .send({ password: "1234" });

  expect(res.statusCode).toBe(400);
});
```

Advanced Bug Detection with Copilot

Prompt Examples

Detect possible null pointer exceptions in this file.
Find security vulnerabilities in this code.
Suggest boundary test cases.
Identify performance bottlenecks.

Bug Detection Workflow with GenAI

```
Code Commit
  ↓
Copilot Review
  ↓
Bug Suggestions
  ↓
Auto Test Generation
  ↓
Developer Fix
  ↓
CI Pipeline Validation
```

Best Practices

Review AI suggestions manually
Add human validation
Combine with static analysis tools
Store detected bug patterns
Train team on prompt writing

Quick Check (Knowledge Test)

1. What is Bug Detection?
2. Name two benefits of GenAI in bug detection.
3. Which prompt helps Copilot find vulnerabilities?
4. Why should return be used after sending response?
5. Give one real-life example where Copilot can detect a bug.